

MARCELO VINICIUS TAPARELLI

AI Engineer | Software Engineer

Python · Machine Learning · RAG · Agents · Evals · TypeScript · PostgreSQL · AWS

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PROFESSIONAL SUMMARY

AI Engineer with a strong Software Engineering foundation and experience in backend, APIs, and production software. I build RAG, agents, and Machine Learning with reliable evaluation and human review. A Product perspective guides approach selection, architecture, security, and executable delivery.

PROFESSIONAL EXPERIENCE

Agência Catus | AI Engineer | Jul 2026 - Present

- Build stores integrated with e-commerce platforms using AI coding agents; maintain Wake implementations and optimize performance/Core Web Vitals.
- Develop and evolve the agency's website and two WordPress blogs, including theme customization, features and server deployment.
- Research AI applications for process automation, development efficiency and product delivery.

EVAG Comunicação e Tecnologia | Software Engineer | Apr 2024 - Present

- Develop and maintain systems with AI coding agents, using Laravel, PHP, Python, JavaScript, WordPress and Tailwind.
- Analyze and develop AI-powered automation to optimize processes and support operations.
- Built Google Drive-to-WordPress automation for petition pages and an internal request management solution integrated with GitLab for traceability.

EVAG Comunicação e Tecnologia | Product Owner | Apr 2023 - Mar 2024

- Managed website delivery from discovery through delivery, connecting clients, design, and development.
- Structured requirements and support workflows, prioritizing across business needs, user experience, and the team's technical capacity.

TECHNICAL SKILLS

AI Engineering: Python, FastAPI, scikit-learn, pandas, Machine Learning, RAG, pgvector, hybrid retrieval, LangGraph, OpenAI, agents, evals, guardrails and human review; experimental model evaluation/adaptation with PyTorch/CUDA.

Software Engineering: TypeScript, Bun, Node.js, REST APIs, PostgreSQL, Redis, Git, Clean Architecture, DDD principles, TDD, unit/integration/E2E tests, idempotency and resilience.

Production, security & reliability: OpenTelemetry, authentication, authorization, performance, Docker, Terraform, CI/CD and a temporary non-production AWS lab.

Product perspective: Problems, requirements, trade-offs, outcomes and deciding when to use AI.

SELECTED PUBLIC PROJECTS

OpsPilot AI: Python/FastAPI, RAG and LangGraph/OpenAI agents with human approval, GitLab reconciliation, evals and OpenTelemetry. Independent OpenAI and GitLab sandbox smokes; no AWS deployment.

Ops Triage AI: Triage with a deterministic baseline, local LLM and human review. Jev/Laya evaluation and PyTorch/CUDA adaptation outside runtime; frozen synthetic held-out of 70 tickets, with documented regressions.

DefectRisk: Python/scikit-learn, risk ranking for human review. Historical test: the top ~30% of modules included 300/421 defective modules (71.26% recall). Leakage-safe evaluation and reproducible CLI; the later artifact needs external validation.

Resilient Transaction API: TypeScript/Bun, PostgreSQL, Redis, idempotency, retries and circuit breaker. Terraform and a temporary non-production AWS lab with functional validation, destroy and audit.

Salus: Node.js/Express, PostgreSQL/Prisma, Clean Architecture, domain modeling and a manual TDD core; JWT, ownership-based authorization and tests.

EDUCATION

Postgraduate Program in AI Engineering — Cruzeiro do Sul

In progress · expected completion: Apr 2027

Systems Analysis and Development — UniBF

Completed in 2026

Software Engineering — Alura

Back-end · Node.js APIs · Authentication · Testing · Security · DevOps/CI/CD · Cloud/AWS · Secure Development

LANGUAGES

Portuguese — native · English — advanced